

High Availability Distributed Systems – Project 1

You are a 2-person team responsible for creating and configuring an enterprise network based on Cisco Packet Tracer, following the guidelines:

1. Design a LAN for a company employing 10 office workers, one of whom works in the HR department, two in accounting, and seven in the software development department.
2. The company is managed by the president, who uses a single computer in his office, and a secretary works in the adjacent office.
3. The company owner wants all employees to have access to a web server and an FTP file server via the company's intranet.
4. Due to company policy, each department is to be located on a separate LAN.
5. Access routers can be used to separate departments.
6. To connect the enterprise network into an intranet, a dedicated network can be used, connecting all routers and the server.
7. In the case of the company's internal network, the addresses of individual subnets can be freely defined.
8. An additional assumption is that computers on a given subnet should communicate with each other and with servers. However, communication with other subnets is to be blocked.
9. Only the president and secretary are to have access to all computers.
10. The secretary is also able to open the gate and turn on/off the lawn watering system in front of the company from her computer.

Project Requirements:

Upload a compressed folder with a name that is a concatenation of the students' surnames (according to the format: LastName1_LastName2) and containing the project.pkt file and documentation in accordance with the 'project_en' template, including the configuration of individual network devices and computers and a cost estimate that includes only the valuation of network devices and materials for a given hardware configuration of the solution.

Project Completion Deadline:

Files must be submitted by May 20, 2026.

Scoring:

A maximum of 15 points can be awarded for correct and complete project implementation.